

Loan Default Risk Analysis

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Description :

Conducted a comprehensive credit risk analysis for Horizon Financial Group to identify key factors contributing to personal loan defaults. Analyzed borrower demographics, income levels, credit scores, employment history, and loan characteristics from over 600 loan applications issued between 2024 and 2025. Evaluated default patterns, borrower risk profiles, and loan performance metrics to uncover trends impacting repayment outcomes. Provided data-driven insights to support improvements to the company's credit scoring model, underwriting process, and loan approval strategies while helping reduce default rates and improve risk management.

Workflow :

1. Imported both data set and review both set look for nulls or any missing data
2. Read each question to make sure I knew what I was looking for
3. Join the data sets on borrower_id
4. Identify import values : debt-to-income (DTI), credit score, and default rate .
5. Look for any correlation between important values and the default rate
6. Built charts showing default rate by credit score range, a scatter plot of DTI vs. default status, and a bar chart comparing loan purposes.

1. What is the overall default rate, and how does it break down by credit score range

The credit score range with the highest default rate is 520–599, with a 49.14% default rate. After looking at the data, you can see a strong correlation between credit score and default rate. The higher the credit score, the lower the default rate will be. One thing that stands out about the data is that the most significant decrease occurs between the 500–599 and 600–649 credit score ranges, where the default rate decreases by 20.11%. However, after reaching the 600–649 credit score range, the decreases become less significant, with only a 1.03% decrease between the 600–649 and 650–699 credit score ranges. The data shows that borrowers with credit scores below 600 represent a much higher risk of default compared to borrowers with higher credit scores.

Credit score ranges:

500 - 599 with a 49.14% default rate
600–649 with a 29.03% default rate
650–699 with a 28.00% default rate
700–749 with a 16.28% default rate
750+ with an 11.69% default rate.

2. Is there a relationship between a borrower's debt-to-income (DTI) ratio and the likelihood of defaulting? What DTI threshold would you recommend as a cutoff for loan approval?

There is a relationship between debt-to-income (DTI) and default rates. If borrowers have a high DTI, they are more likely to default. The cutoff for loan approval should be set at 50% because when borrowers have a DTI higher than 50%, that is when you see a lot of borrowers defaulting.

3. Which loan purposes have the highest default rates, and does the average loan amount differ significantly between defaulted and non-defaulted loans?

The top 3 loan purposes with the highest default rates are wedding loans with a 32.14% default rate, home improvement loans with a 28.57% default rate, and auto loans with a 27.12% default rate. There is a significant difference between defaulted and non-defaulted loans, but when looking at each loan purpose, you will notice that non-defaulted loans are higher for each loan purpose except for education, auto loans, debt consolidation, and medical expenses.

4. How do employment status and years employed affect default risk? Are borrowers with less than 2 years of employment significantly more likely to default?

Yes, borrowers who have worked part-time and full-time but have been employed for less than two years have a higher default rate. This would suggest that if you want borrowers to be less likely to default, they should be employed for more than two years, especially if they are working part-time.